Lab 4 | Phạm Nguyễn Công Danh | ITDSIU23035 | IT171IU – Lab Group 01

Exercise 1

Interpretation

(c)

```
Seed 0 = np.float64(0.029),
Seed 1 = np.float64(0.025),
Seed 2 = np.float64(0.0248)
```

The validation error rate fluctuates slightly depending on the random seed (approximately 2.48% to 2.9%). This highlights a limitation of the validation-set approach: the estimated result depends on which data points happen to be randomly assigned to the training or validation sets.

(d)

```
Validation error when add student variable:  [np.float64(0.0254)]
```

Model with student: Adding the student variable does not significantly increase the validation error rate Consequently, incorporating the student variable into the model does not improve or reduce its actual predictive performance.

Exercise 2:

(a) standard errors of the income and balance

```
income     0.000005
balance    0.000227
```

(b)

```
intercept   -11.540468
income        0.000021
balance       0.005647
```

(c)

```
intercept    0.435692
income       0.000005
balance      0.000230
```

(d)

The bootstrap standard error estimates (0.000005 for income and 0.000230 for balance) are extremely close to the formula-based standard errors obtained from sm.GLM (0.000005 for income and 0.000227 for balance).

The formula-based standard errors are highly trustworthy in this case, indicating that the underlying assumptions of the logistic regression model hold well for the Default dataset.

Exercise 3

```
LOOCV error rate: 0.44995408631772266
Baseline error rate (Always Up): 0.4444444444444444
```

Analysis: Fitting a logistic regression model of Direction on Lag1 and Lag2 yields a LOOCV test error rate of 44.99%. This is slightly worse than the naive baseline model—which blindly predicts "Up" for every observation—with an error rate of 44.44%.

Conclusion: The lag variables Lag1 and Lag2 possess virtually no predictive power for market direction. The model fails to outperform a simple majority-class guessing strategy.

Exercise 4

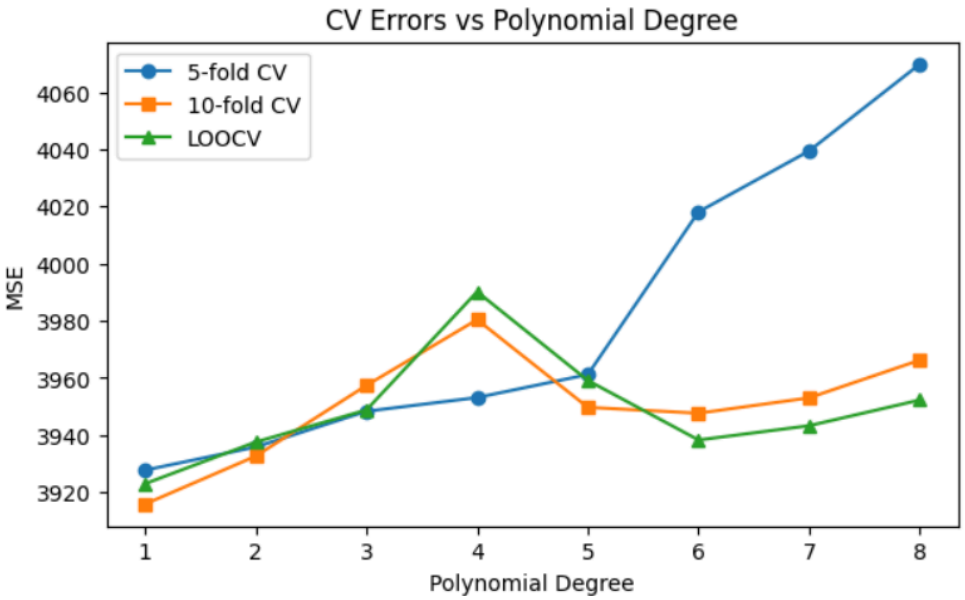
```
(c) LOOCV Errors (Seed 1): [6.63302984 1.12293686 1.30179655 1.33239427]
(d) LOOCV Errors (Seed 99): [4.58971213 1.17353228 1.22513802 1.32238896]
```

Model & Data Description: The simulated dataset is generated from $Y = X - 2X^2 + \varepsilon$ with $n = 100$ observations and $p = 2$ effective predictor terms. The scatter plot reveals a clear downward parabolic trend.
CV Error Analysis: Across both random seeds, the LOOCV MSE drops dramatically from the linear model ($d = 1$, MSE $\approx 4.59 - 6.63$) to the quadratic model ($d = 2$, MSE $\approx 1.12 - 1.17$). Increasing polynomial degrees to $d = 3$ and $d = 4$ leads to slightly higher test errors due to overfitting.
Conclusion: The quadratic model ($d = 2$) minimizes the LOOCV test error, which perfectly matches the true underlying data generation process ($Y = X - 2X^2 + \varepsilon$). Furthermore, LOOCV results for a fixed dataset are deterministic and do not vary between runs.
Exercise 5

```
22.532806324110677 0.40886114749753505
Bootstrap SE for Mean: 0.4127
(c) Bootstrap 95% CI: [21.7073, 23.3583]
    t-test 95% CI:    [21.7294, 23.3362]
(d) Median: 21.2000, Bootstrap SE: 0.3780
    10th Percentile: 12.7500, Bootstrap SE: 0.4978
```

Standard Error & Confidence Interval Comparison: The bootstrap standard error estimate ($SE = 0.4127$) closely matches the formula-based standard error ($SE = 0.4089$). The 95% bootstrap confidence interval ($[21.7073, 23.3583]$) is virtually identical to the standard t-test interval ($[21.7294, 23.3362]$).
Other Statistics: The sample median of medv is 21.2000 with a bootstrap SE of 0.3780. The 10th percentile is 12.7500 with a bootstrap SE of 0.4978.
Conclusion: The bootstrap provides an accurate standard error estimate for the mean (matching classical theory) while effortlessly extending to non-standard statistics like medians and percentiles that lack simple analytical formulas.
Exercise 6

...




```
10-fold CV errors: [np.float64(3915.860520587423),
np.float64(3932.665129215356), np.float64(3957.399772144444),
np.float64(3980.4109154984703), np.float64(3949.717867056321),
np.float64(3947.6596958489085), np.float64(3953.044685983006),
np.float64(3966.2538499710126)]
```

Curve Analysis: The 10-fold cross-validation MSE achieves its minimum at degree 1 ($d = 1$) with an MSE of 3915.86. As the polynomial degree increases, test error consistently grows due to increased model complexity and overfitting on single-variable polynomial terms.

One-Standard-Error Rule: Applying the 1-SE rule points directly to the simplest model within one standard error of the minimum MSE, which is the degree-1 linear fit.

Conclusion: A simple linear model ($d = 1$) is preferred for predicting progression from bmi. Adding higher-degree polynomial terms increases variance without improving test set prediction accuracy.